

Neuroscientific Frontiers — Module 08: Planning —

Study Questions

1. What is computational psychiatry? How does Active Inference provide a unified framework for understanding psychiatric symptoms as inference failures?
2. What are the three levels of inferential failure in psychiatry (precision failure, model misspecification, structure failure)? Give a clinical example of each.
3. Explain the Active Inference model of hallucinations in schizophrenia. How does excessive sensory precision lead to interpreting noise as signal?
4. How do delusions form under aberrant precision? What is the "jumping to conclusions" bias and how does Active Inference formalize it?
5. How are negative symptoms of schizophrenia (flat affect, avolition, social withdrawal) explained within Active Inference? Are they a primary precision deficit or a secondary compensatory mechanism?
6. Explain the dopamine hypothesis of schizophrenia in Active Inference terms. How do antipsychotics work by reducing precision of prediction errors?
7. How does Active Inference model depression as inference under persistently pessimistic priors? What is the role of the C vector in maintaining negative cognitive bias?
8. What is learned helplessness in Active Inference terms? How does repeated negative experience reduce policy precision and produce passivity?
9. How does allostatic overload contribute to depression? What does it mean for homeostatic setpoints (C vector) to shift to accommodate chronic stress?
10. Explain panic disorder as catastrophic interoceptive inference. Why does a minor heart rate fluctuation trigger a panic attack in susceptible individuals?
11. How does the locus coeruleus norepinephrine system contribute to anxiety disorders? What does "overestimation of environmental volatility" mean clinically?
12. What is the HIPPEA hypothesis of autism? How does inflexibly high precision on prediction errors explain enhanced detail perception, sensory overwhelm, and insistence on sameness?
13. How does Active Inference explain the social communication difficulties in autism? Why might social signals specifically require flexible precision?

14. How does addiction hijack policy precision? What happens to the policy space as the addicted model progressively narrows its repertoire of evaluated policies?
15. What is withdrawal as prediction error? Why does the fastest resolution of withdrawal prediction errors involve taking the substance again?
16. Compare the Active inference model of addiction with incentive salience theory (Berridge & Robinson). Where do they converge and diverge?
17. How might computational phenotyping (Research Methods Module 02) be used to personalize psychiatric treatment based on an individual's specific inferential parameters?
18. What is the role of placebo responses in Active Inference psychiatry? Are placebos "real" treatments that change priors, or mere expectation effects?
19. How might psychedelic-assisted therapy work through Active Inference? What does 5-HT_{2A}-mediated precision relaxation do to rigid pathological priors?
20. Critically evaluate: Does computational psychiatry through Active Inference genuinely improve on symptom-based (DSM) diagnosis, or does it simply re-describe the same phenomena in mathematical language? What clinical trial would prove it adds value?